BUDGET MEMO

Drive models, replacement age, and spares for the 2027 buy

To: Infrastructure Director
From: Reliability analysis
Date: 23 October 2026
Re: What to buy, when failure risk rises, and how many spares to hold

Standardize the 2027 capacity buy on WDC WUH722222ALE6L4 (WDC, 22 TB, AFR 0.47% on 282 failures). Keep TOSHIBA MG11ACA24TE as the second source (0.50% on 23 failures), at 20% of terabytes. Stop buying 13 current models whose lower 95% bound is above the fleet rate, led by ST10000NM0086 at 6.08%, including the 24 TB Seagate ST24000NM002H at 3.04%. The age 1–3 hazard is 0.68% per year. It first doubles that rate at 4.0 power-on years (1.41% per year). It keeps climbing, to 7.14% per year at 8.5 years.

1. What to buy, and what to stop buying

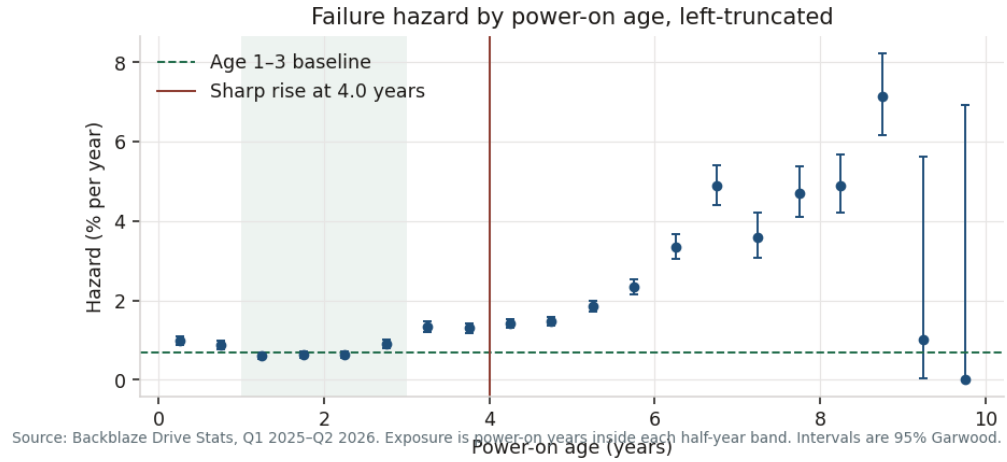
Hard-drive AFR from 2025-01-01 to 2026-06-30 is 1.40% (95% interval 1.37% to 1.44%), on 6,861 failures and 178,372,531 drive-days. A model is judged with at least 200 drives, 100,000 drive-days, 8 TB, and a drive still present in 2026Q2. Stop means the lower 95% bound is above the fleet rate. A buy candidate's point AFR also has to stay within 25% of the fleet. The cost screen's cheapest approved model was TOSHIBA MG11ACA24TE. The cheapest approved model has 23 failures and is not in every large datacenter. WDC WUH722222ALE6L4 has 282 failures and is the 80% buy.

Model	TB	AFR	95% interval	Drives	Decision
WDC WUH722222ALE6L4	22	0.47%	0.42%–0.53%	45,893	Standardize
TOSHIBA MG11ACA24TE	24	0.50%	0.31%–0.74%	9,630	Second source
ST10000NM0086	10	6.08%	4.89%–7.47%	1,044	Stop buying
ST14000NM0138	14	5.72%	4.69%–6.91%	1,322	Stop buying
TOSHIBA MG08ACA16TEY	16	5.14%	4.64%–5.68%	5,197	Stop buying
ST12000NM0007	12	5.06%	3.98%–6.34%	1,038	Stop buying
HGST HUH721212ALN604	12	4.54%	4.20%–4.89%	10,195	Stop buying
ST24000NM002H	24	3.04%	2.71%–3.41%	9,900	Stop buying

The table shows the six highest stop-list rates. 7 more stops are on the dashboard.

2. When failure risk rises

The age 1–3 hazard is 0.68% per year. It first doubles that rate at 4.0 power-on years (1.41% per year). It keeps climbing, to 7.14% per year at 8.5 years. The first half-year is also elevated, at 0.98% per year, so new drives need spares before wear-out begins.



Age is power-on years from SMART 9, not the date a drive first appears in the file. Drives that are already old on their first snapshot are left-truncated: they count only from the age at which we can see them.

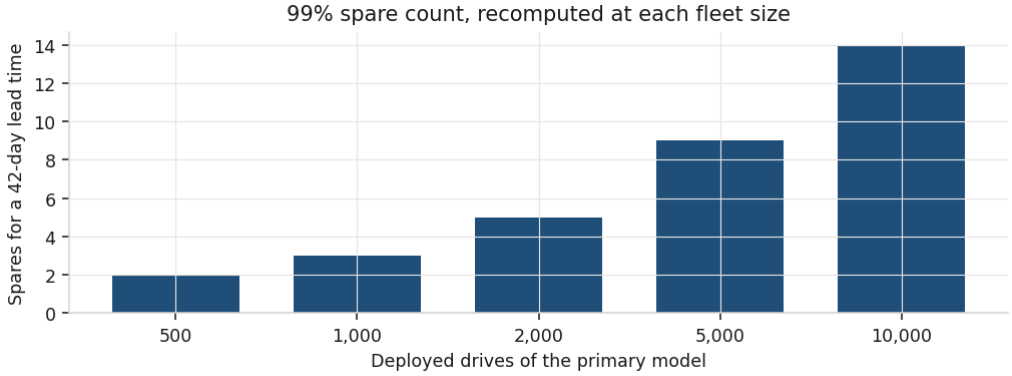
3. How many spares to hold

For a 100 PB raw refresh, buy 80% of the terabytes as WDC WUH722222ALE6L4 (3,637 drives) and 20% as TOSHIBA MG11ACA24TE (834 drives). Stock 7 and 3 spares for a 42-day lead time at a 99% service level. The spare buy is $3,164, inside the $100,000 spare budget. With an empty shelf, expected shortfall over that lead time is 68 TB.

Spares for WDC WUH722222ALE6L4 use its age 0-1 hazard (0.62%), against an all-age AFR of 0.47%. Planning prices are about $14 to $22 per TB, lower for larger drives, and are not quotes. The mission is 100 PB raw and the spare budget is $100,000.

Single-model spare counts for the primary drive. Each row is recomputed.

Deployed drives	Expected failures in 42 days	Spares at 99%	Shortfall with zero spares
500	0.4	2	8 TB
1,000	0.7	3	16 TB
2,000	1.4	5	31 TB
5,000	3.6	9	78 TB
10,000	7.1	14	157 TB



Each bar is a Poisson quantile at that fleet size. It is not a linear scaling of the 1,000-drive count.

Recommended primary-model quantity, 99% service: 21 days needs 5; 42 days needs 7; 84 days needs 11.

Recommendation

• Approve WDC WUH722222ALE6L4 for 80% of new terabytes and TOSHIBA MG11ACA24TE for 20%.

• Stock 7 WDC WUH722222ALE6L4 spares and 3 TOSHIBA MG11ACA24TE spares (42 days, 99% service).

• Stop new purchases of 13 models, listed on the dashboard. Do not buy ST24000NM002H as a 24 TB alternative.

• Put drives at or above 4.0 power-on years on the replacement list. Use power-on age, not the date the drive first appears in a report.

• If the fleet or the lead time changes, recompute the Poisson quantile at the new size. Do not scale the spare count by dividing.

What this does not tell you

• These are Backblaze data-center drives under Backblaze's workload and replacement practice. The rates describe that fleet. They are not a lab test, and they are not a claim about another company's rooms, temperature, or duty cycle.

• Model differences are associations. Backblaze chooses which model goes into which pod. Datacenter annualized rates differ by enough that model and location are mixed together.

• The age-scale Cox model cannot take a Schoenfeld test in this software once left truncation is included. A second Cox model, on time since the first snapshot, failed the proportional-hazards test, so those hazard ratios are not used. The buy list comes from the annualized rates and the life table.

• A drive that disappears without failure = 1 is treated as censored, which is right if it was migrated and wrong if the death was never flagged. SMART 9 units were checked per model; drives with an unusable counter are in the AFR and out of the age curve.

• The 100 PB mission, the 42-day lead time, the 99% service level, the $100,000 spare budget, and the planning prices are assumptions for this fictional decision. Change one of them and the spare count changes. The reliability ranking does not depend on the prices.

Method, in one line: AFR = failures / drive-days × 365, with Garwood 95% intervals. Age hazard is a half-year life table with left truncation. Spares are the smallest Poisson count whose CDF is at least 99% at mean = drives × rate × 42/365. Source: Backblaze Drive Stats, Q1 2025–Q2 2026, downloaded 23 October 2026 from the Backblaze hard drive test data page. Raw files are not in this repository. Definitions and the run order are in the README.